PRIVACY-PRESERVING MACHINE LEARNING FOR COLLABORATIVE DATA SHARING VIA AUTO-ENCODER LATENT SPACE EMBEDDINGS

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INTRODUCTION

Privacy-preserving machine learning in data-sharing processes is an ever-critical task that enables collaborative training of Machine Learning (ML) models without sharing the original data sources. It is especially relevant when an organization must ensure that sensitive data remains private throughout the ML pipeline, i.e., training and inference phases. This paper presents an innovative framework that uses Representation Learning via autoencoders to generate privacy-preserving embedded data. Thus, organizations can share the data representation to increase machine learning models' performance in scenarios with more than one data source for a shared predictive downstream task.

PROPOSED COLLABORATIVE FRAMEWORK

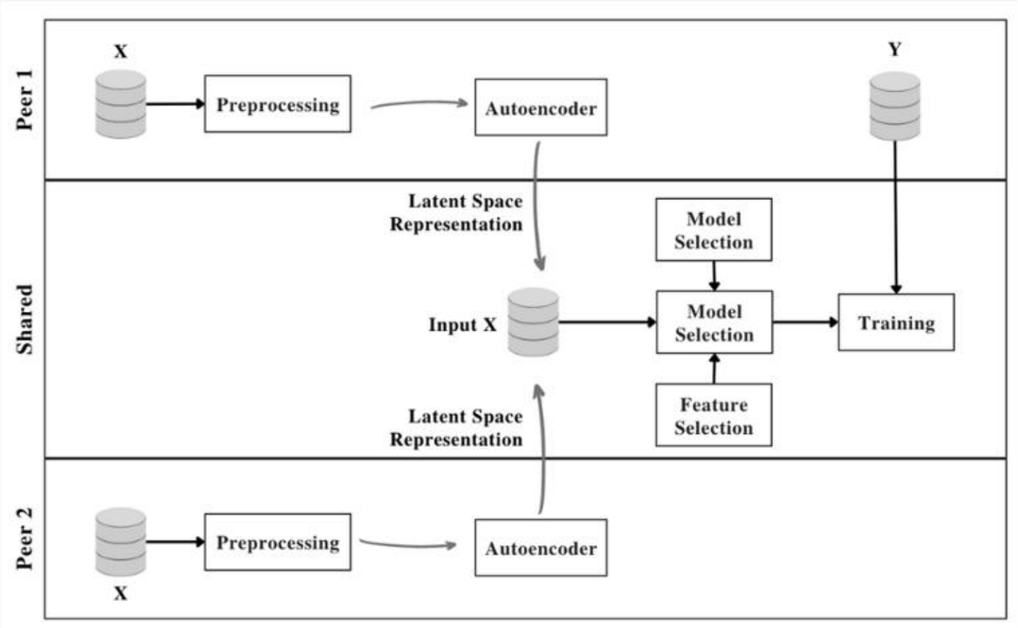


Figure 1: Privacy-Preserving Machine Learning Framework: the two peers share the respective latent representation of the corresponding database to a shared data source that is then used to train the Machine Learning model in the training phase.

BASE AUTOENCODER SETUP

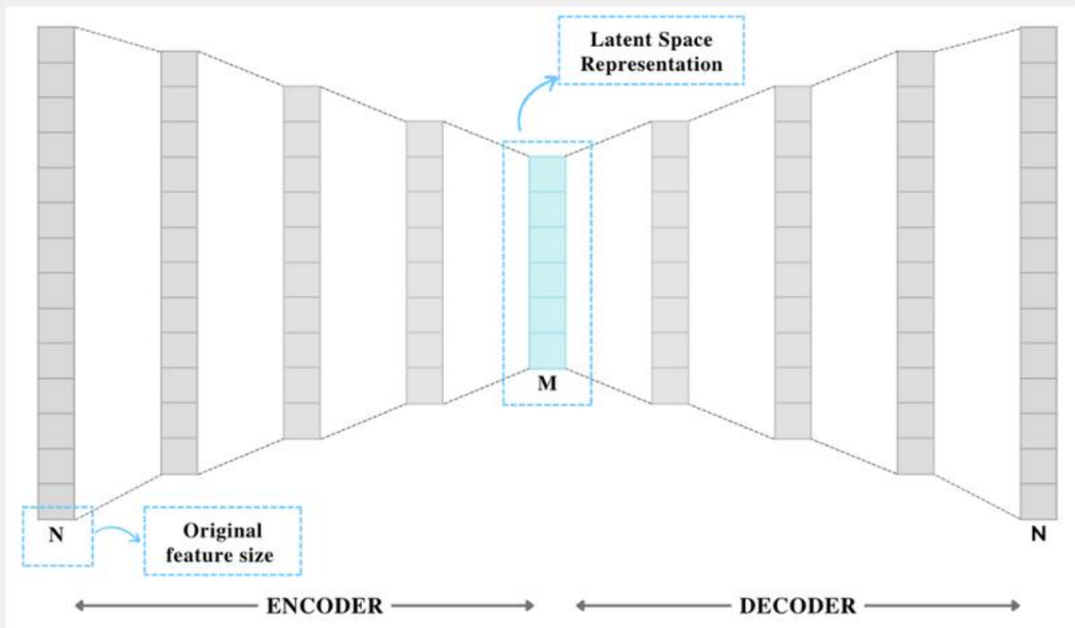


Figure 2: The Autoencoder presents two components: encoder and decoder, each have four layers, and the connection between both is the latent space representation layer. N represents the original feature size, and M the embedding size.

EXPERIMENTAL SETUP

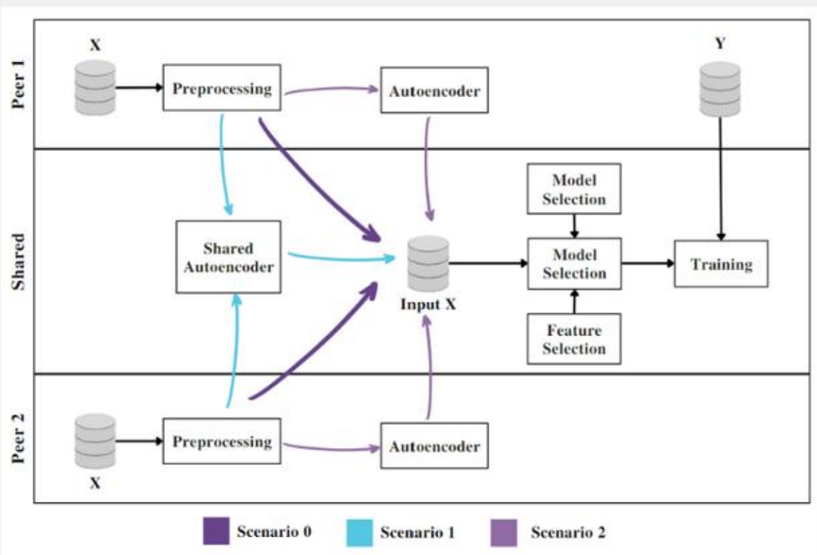
Scenario 0: Predictive model for the downstream task using a single data source

Scenario 1: Unique dataset to obtain a single representation vector and use it to train a predictive model

Scenario 2: Simulate two peers, preprocess them individually and obtain a representation vector.

Scenario 3 and 4: Transform the base autoencoder into a multitask neural network that, on one side, predicts the representation and, on the other, predicts the objective variable. Replicate the structure of scenarios 1 and 2, respectively

Figure 3: Presents the pipeline representation of scenarios 0, 1, and 2.



RESULTS

The following results correspond to test the scenarios in 3 case studies

House Prices

Observations: 21,613

Features: 12

ML Task: Regression

Representation Error: 5% - 10%

	Metrics	Scenario 0	Scenario 1	Scenario 2	Scenario 3	Scenario 4
Train	R2	96.17%	91.13%	89.28%	94.21%	89.01%
	MAPE	22.25%	28.46%	32.18%	25.38%	30.27%
Validation	R2	96.14%	91.08%	89.68%	93.87%	88.47%
	MAPE	24.76%	28.51%	33.42%	26.19%	31.06%
Test	R2	96.19%	91.55%	89.01%	94.03%	87.90%
	MAPE	23.87%	28.94%	33.23%	25.94%	31.89%

Mnist Numbers

Observations: 35,000

Features: 784

ML Task: Multiclass Classification

Representation Error: 2% - 8%

	Metrics	Scenario 0	Scenario 1	Scenario 2	Scenario 3	Scenario 4
Train	Accuracy	94%	88%	84%	92%	85%
	Precision	94%	88%	84%	92%	85%
	Recall	94%	88%	84%	92%	85%
Validation	Accuracy	92%	88%	84%	91%	84%
	Precision	92%	88%	84%	91%	84%
	Recall	92%	88%	84%	91%	84%
Test	Accuracy	92%	88%	84%	91%	84%
	Precision	92%	88%	84%	91%	84%
	Recall	92%	88%	84%	91%	84%

Buzz in Social Media

Observations: 87,488

Features: 12

ML Task: Regression

Representation Error: 6% - 11%

	Metrics	Scenario 0	Scenario 1	Scenario 2	Scenario 3	Scenario 4
Train	R2	90.26%	84.26%	89.30%	89.79%	88.78%
	MAPE	15.31%	18.03%	16.03%	15.69%	17.74%
Validation	R2	90.32%	84.41%	89.30%	88.93%	87.26%
	MAPE	14.88%	17.91%	15.89%	15.39%	16.89%
Test	R2	90.29%	84.27%	89.33%	89.21%	87.36%
	MAPE	15.09%	17.97%	15.96%	15.27%	17.58%

CONCLUSIONS

Our results prove that with an accurate representation learning model, peers can share an embedded dataset that follows the observations' patterns and behavior. Changing the original features to a latent space representation does not drastically deteriorate the performance of the downstream task. Therefore, peers or organizations can collaborate without risking the organization's privacy policies or violating potential clients' privacy concerns.

FUTURE WORK

Each data source should develop a custom autoencoder neural network implementation to improve the representation performance. In addition, even if we assume that dimensionality reduction keeps data privacy, we will establish measurements to quantify the privacy level for each dataset. This measure should take into account the complexity of the embedding and the difficulty of decoding the original dataset.

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